



PL-7008:

Create agents in Microsoft Copilot Studio

Module 3: Work with entities and variables in Microsoft Copilot Studio



Microsoft Learn module

Work with entities and variables in Microsoft Copilot Studio

<https://learn.microsoft.com/training/modules/power-virtual-agents-entities/>



Learning Objectives

After completing this module, you will be able to:

- 1** Work with entities
- 2** Create custom entities
- 3** Configure variables

Work with entities



Entities

City

Description

City names, extracted as a string

Method

Prebuilt

Usage

Defines the extracted text for entity.

User input	Entity	Saved value
The business headquarters is in Seattle	Seattle	Seattle
I love NYC	NYC	New York City
I'm going to Vancouver B.C.	Vancouver B.C.	Vancouver B.C.

Shows the saved value stored in variable.

Prebuilt entities

Commonly used information

- Person
- Location
- Numbers
- Date and times
- Colors

Settings

Generative AI

Agent details

Security

Connection Settings

Authoring Canvas

Entities

Skills

Voice

Languages

Language understanding

Component collections

Advanced

+ Add an entity

Name	Description	Method
Age	Age of a person, place, or thing, extracted as a number	Prebuilt
Boolean	Positive or negative responses, extracted as a Boolean	Prebuilt
City	City names, extracted as a string	Prebuilt
Color	Primary colors and hues on the color spectrum, extracted as a string	Prebuilt
Continent	Continent names, extracted as a string	Prebuilt
Country or region	Country and region names, extracted as a string	Prebuilt
Date	Dates, days of the week, and months relative to a point in time, extracted as a string	Prebuilt
Date and time	Dates, times, days of the week, and months relative to a point in time, extracted as a string	Prebuilt
Date and time without timezone	Dates, times, days of the week, and months relative to a point in time, extracted as a string without the timez...	Prebuilt
Duration	Lengths of time, extracted as a string, in standard TimeSpan format	Prebuilt

Using an entity

Question node

- Select an entity for Identify
- Agent smartly recognizes an entity and extracts the relevant information from the user input and save it to a variable



Question

Text ☐ 

What city do you live in?

Identify

☐ City 

Save response as

 {x}

UserCity

string



Using a closed list entity

Question node

- Choose a closed list entity for Identify
- Must select the list values from entity

The screenshot shows a user interface for selecting a property type. The main window has a title bar and a content area. The content area contains a question "What type of property do you want to..." and an "Identify" section with a dropdown menu labeled "Property Type". Below the dropdown is a link "Select options for user". Underneath is a section "Options for user" with four buttons: "Apartment", "House", "Condominium", and "Duplex". A modal window titled "User options for Property Type" is open on the right. It has a search bar and a table with four rows, each representing a property type and its selection status.

User options for Property Type	
List values (4)	
	Display
Apartment	☒
House	☒
Condominium	☒
Duplex	☒

Custom entities



Create a custom entity

Entity types

- Closed list
- Regular expression

Create an entity



Make your copilot smarter with entities.

An entity is a piece of information (person, place, thing, or concept) that the copilot might want to pick out from what your customer says.

[Learn how to use entities](#)

Choose a method

Closed list

Define a list. Use when you have an exact list of items that is a manageable size (small to medium).

Example: List of sizes

- Extra small
- Small
- Medium
- Large

Regular expression (Regex)

Define a specific logical pattern, such as an ID number, credit card number, IP address, and more.

Example: Article ID pattern that starts with "kb" and ends with 6 digits, such as 'kb123456'.

`kb[0-9]{6}`

Closed list entity

List of items

- Add names
- Add synonyms

Outdoor Store Categories

Name *

Outdoor Store Categories

Description

Description (optional)

Method

List

The bot will try to match an item on the list based on what the customer says.

Modified by

2 minutes ago

Smart matching

☒ on

The Smart matching option enables the bot's understanding of natural language. This can help match misspellings, grammar variations, and words with similar meanings.

If the bot isn't matching enough related words, enhance the bot's understanding

List items

Enter item Add

Item	Synonyms
Baseball	+ Synonyms
Basketball	+ Synonyms
Biking	+ Synonyms
Bowling	+ Synonyms
Camping	+ Synonyms
Hiking	+ Synonyms
Cricket	+ Synonyms

Save Close

These items will be used as values stored in variables.

Regular expression entity

Regex

- Specify a regex pattern
- .NET regular expressions syntax
- Use cases
 - Find specific character patterns
 - Validate text to ensure that it matches a predefined pattern
 - Extract, edit, replace, or delete text substrings

Regex selection

Name *

Regex selection

Description

This is a regex to match on user selections.

Pattern *

Enter regular expression pattern

The pattern of a regex type cannot be empty

Method

Regular expression (Regex)
The copilot will extract any text that matches the specific pattern.
[Learn how to use entities](#)

Examples

Knowledge base article ID that starts with "kb" and ends with 6 digits, "kb123456" for example.
kb[0-9]{6}

Credit card number that has 13 or 16 digits and starts with a 4.
4[0-9]{12}(?:[0-9]{3})?

Tracking number that starts with 9 digits and ends with one letter (upper or lower case), "100456789A" for example.
(?i)\d{9}[A-Z]{1}

Save

Close

Demo - Create a closed list entity



Demo

Example: Closed list entity with synonyms

Property Type

Name *

Property Type

Description

Description (optional)

Method

List

The copilot will try to match an item on the list based on what the customer says.

Smart matching

on

List items

Enter item

Add

Item	Synonyms
Apartment	Flat
House	Single-family home
Condominium	Townhouse
Duplex	+ Synonyms

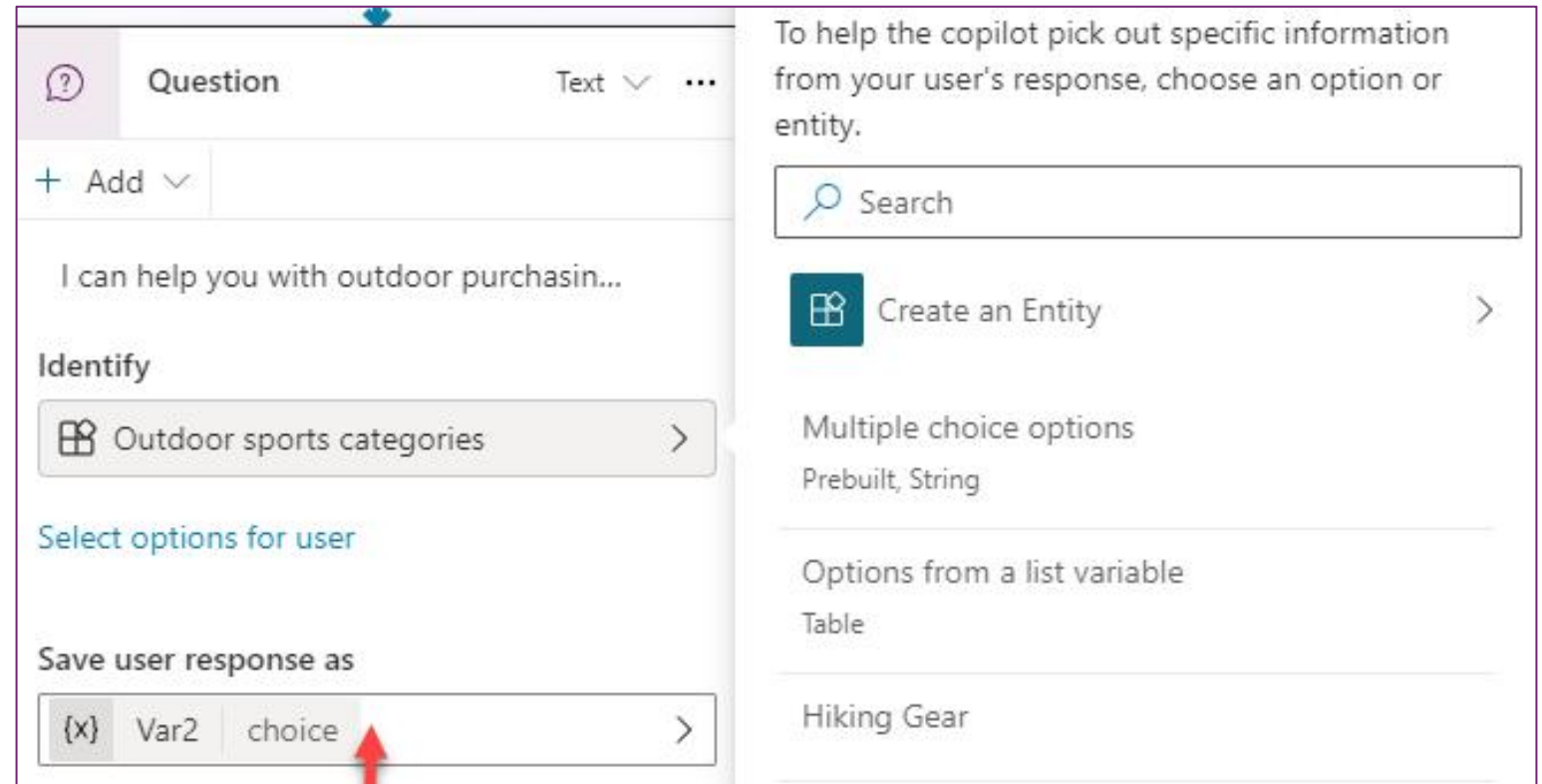
Variables



Create variables

Create in Question node

- Created automatically
- Rename



The screenshot displays the configuration interface for a 'Question' node in a bot. The main panel on the left shows the question text 'I can help you with outdoor purchasin...' and the 'Identify' section with 'Outdoor sports categories' selected. The 'Save user response as' section is highlighted with a red arrow, showing the variable name '{x} Var2' and the choice 'choice'. The right sidebar provides instructions on how to help the copilot pick out specific information from the user's response, offering options like 'Create an Entity', 'Multiple choice options', and 'Options from a list variable'.

Question Text ...

+ Add

I can help you with outdoor purchasin...

Identify

Outdoor sports categories

Select options for user

Save user response as

{x} Var2 choice

To help the copilot pick out specific information from your user's response, choose an option or entity.

Search

Create an Entity

Multiple choice options
Prebuilt, String

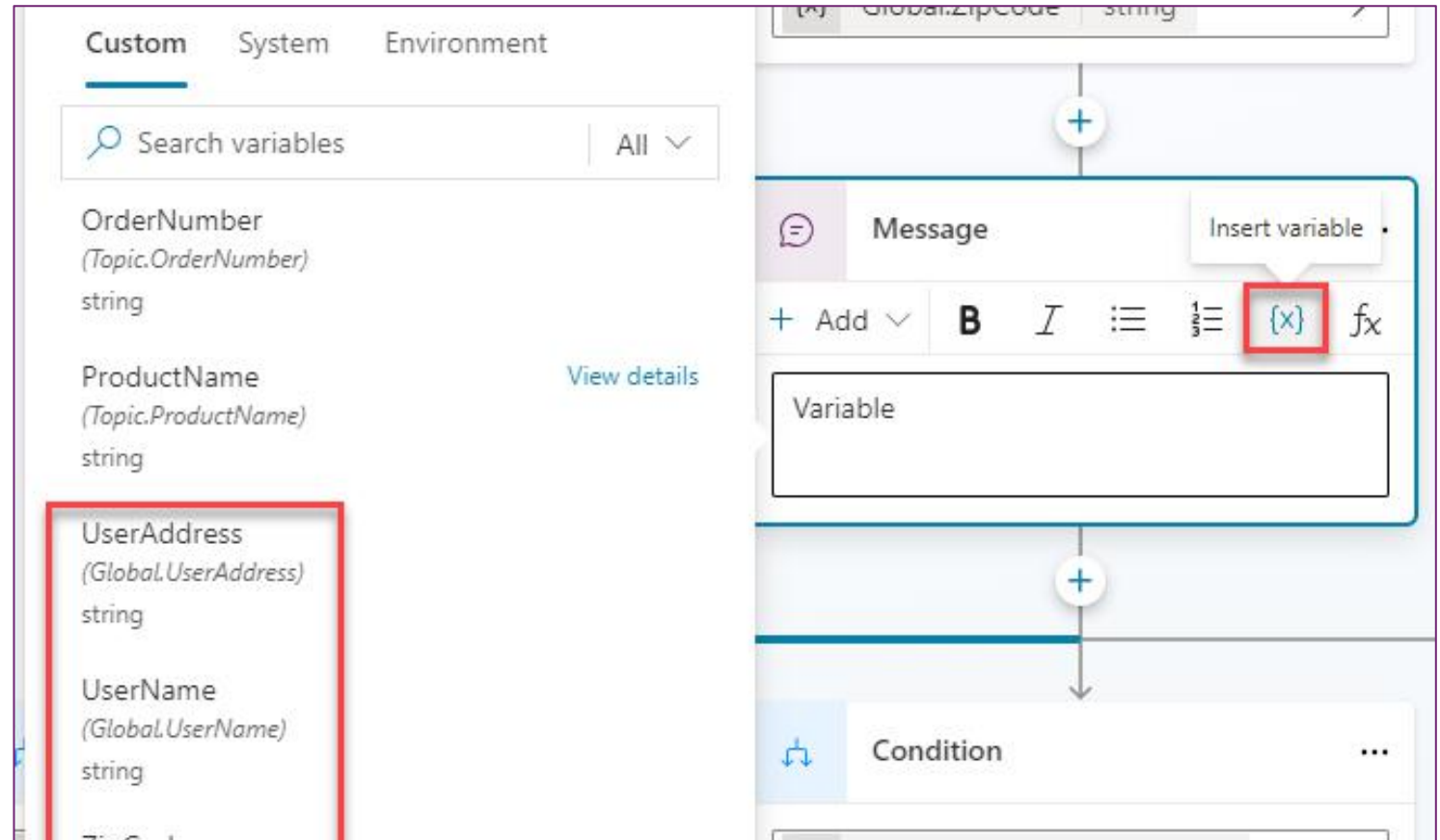
Options from a list variable
Table

Hiking Gear

Use variables

{x} icon

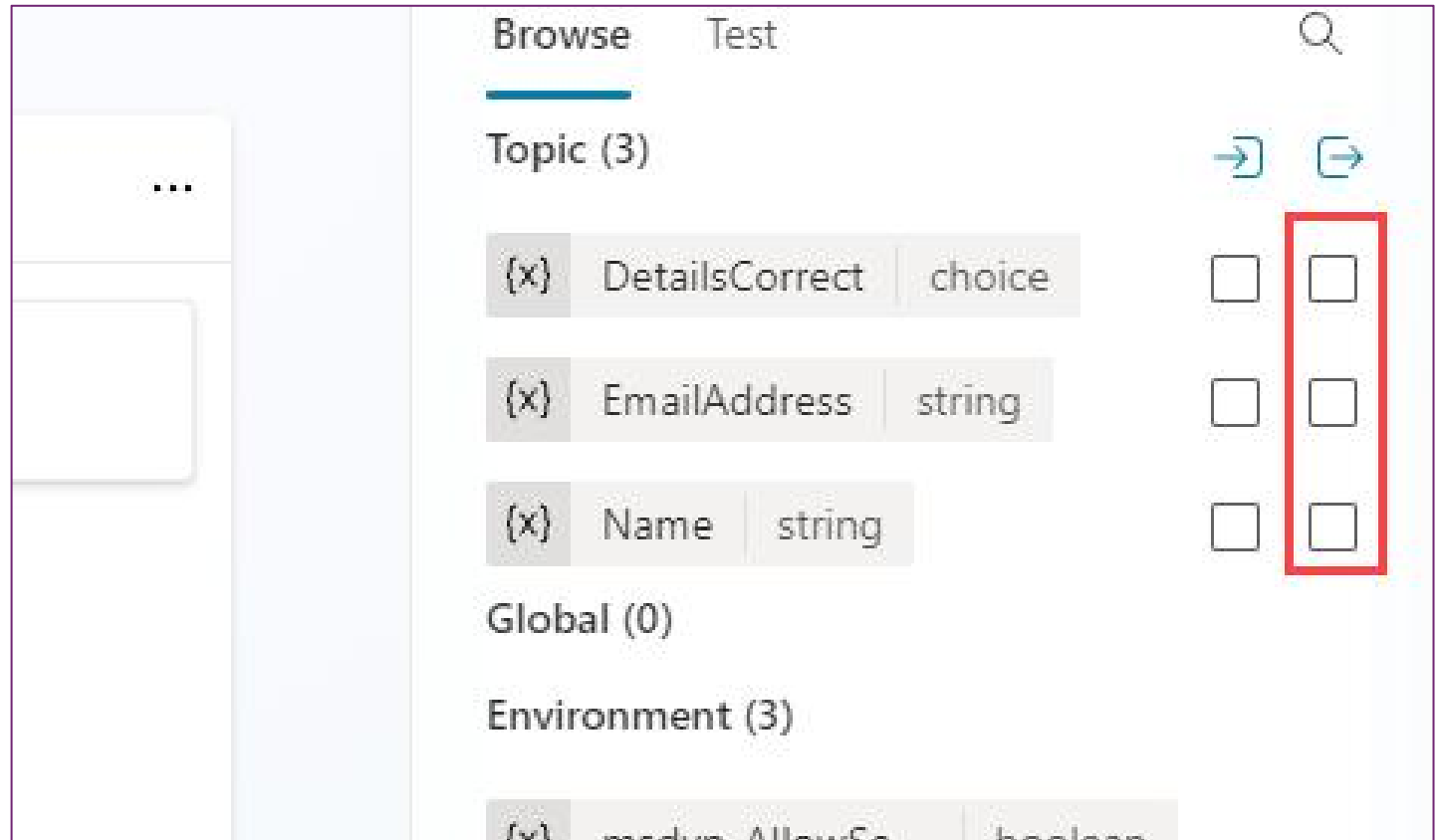
- Insert variable into a node



Pass variables between topics

Variable properties

- Receive values from other topics
- Return values to original topics



Variable scope

Variable types

- Topic (limited scope)
- Global

Variable properties

×

Variable name

DetailsCorrect

Type

choice

Reference

 Question
Are the details correct?

⋮

[View all references](#)

Usage

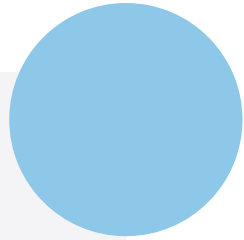
☒ Topic (limited scope)

☐ Receive values from other topics

☒ Return values to original topics

☐ Global (any topic can access)

Practice Labs

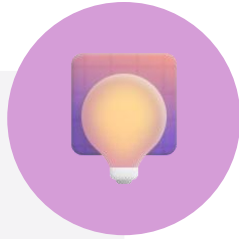


Lab 6: Work with entities

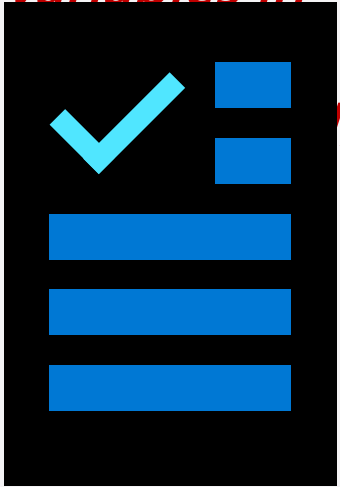
Knowledge Check



Knowledge Check



*Work with
entities and
variables in*



lio

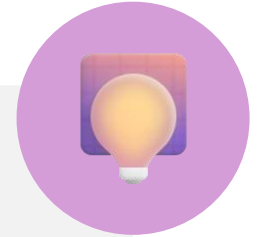
Module 3

1. Use your PC or mobile device
2. Go to (*insert polling app link of your choice*)
3. Enter code: **123-45-678**
4. Please participate

Which one?

- A. None
- B. A little
- C. Lots

Knowledge Check



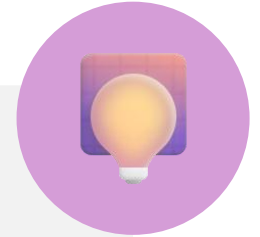
1 You want to create a container to store custom product types that can be used as responses to customer questions, and variables that can be used throughout your agent. What Copilot Studio feature, should you use?

- ✓ Actions
- Entities
- Topics
- Variables

Solution

Answer=B

Knowledge Check



2 You're creating a customer entity that will represent different departments your organization has. Your Sales and Marketing departments work closely together. You want to ensure that if someone indicates they have a Marketing question, that they're being sent to the Sales topic. How would you do this?

- ✓ Add Marketing as a Synonym of Sales
- Add Sales as a Synonym of Marketing
- Make Marketing a child of Sales
- Make Sales a child of Marketing

Solution

Answer=A

Knowledge Check



3 Which of the variables defined below would be available to use across all topics in an agent?

all.UserName

copilot.UserName

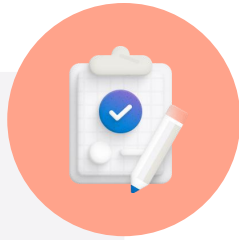
✓ global.UserName

pva.UserName

Solution

Answer=C

Summary



- Entities are used to assist the agent in determining the intent of a user's response
- Entities reduce the need for topics
- You can create closed list and regular expression custom entities
- Variables are created from user responses and can be used to control the conversation flow and passed to actions

Learning Path Recap

In this learning path, we:

Learned how to use entities in question nodes in topics

Learned how to create custom entities

Learned how to create and use variables

